



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Fundamentals of Data Engineering
Course Number:	COE848
Semester/Year (e.g. F2016)	W2026

Instructor:	Dr. Yasaman Ahmadiadli
Lab Report Number	Lab 3
Lab Title	Lab 3: Database Design

Due Date	February 26, 2026
Submission Date	February 26, 2026

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PRAGMA foreign_keys=OFF;
BEGIN TRANSACTION;
CREATE TABLE Restaurant (
    restaurant_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    address TEXT NOT NULL,
    phone_number TEXT NOT NULL,
    cuisine_type TEXT,
    opening_hours TEXT,
    Closing_hours TEXT,
    total_capacity INTEGER
);
CREATE TABLE Customer (
    customer_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    email TEXT UNIQUE,
    phone_number TEXT,
    loyalty_status TEXT
);
CREATE TABLE DiningTable (
    restaurant_id INTEGER NOT NULL,
    table_number INTEGER NOT NULL,
    seating_capacity INTEGER NOT NULL,
    availability_status TEXT,
    seating_location_description TEXT,
    PRIMARY KEY (restaurant_id, table_number),
    FOREIGN KEY (restaurant_id)
        REFERENCES Restaurant(restaurant_id)
        ON DELETE CASCADE
);
CREATE TABLE Orders (
    order_id INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id INTEGER NOT NULL,
    restaurant_id INTEGER NOT NULL,
    reservation_id INTEGER,
    order_date TEXT NOT NULL,
    total_amount REAL,
    order_status TEXT,
    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id),
    FOREIGN KEY (restaurant_id)
        REFERENCES Restaurant(restaurant_id),
    FOREIGN KEY (reservation_id)
        REFERENCES Reservation(reservation_id)

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);
CREATE TABLE Staff (
    staff_id INTEGER PRIMARY KEY AUTOINCREMENT,
    restaurant_id INTEGER NOT NULL,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    role TEXT,
    hourly_wage REAL,
    FOREIGN KEY (restaurant_id)
        REFERENCES Restaurant(restaurant_id)
);
CREATE TABLE MenuItem (
    menu_item_id INTEGER PRIMARY KEY AUTOINCREMENT,
    staff_id INTEGER NOT NULL,
    item_name TEXT NOT NULL,
    price REAL NOT NULL,
    availability_status TEXT,
    FOREIGN KEY (staff_id)
        REFERENCES Staff(staff_id)
);
CREATE TABLE OrderHandledBy (
    order_id INTEGER NOT NULL,
    staff_id INTEGER NOT NULL,
    PRIMARY KEY (order_id, staff_id),
    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id),
    FOREIGN KEY (staff_id)
        REFERENCES Staff(staff_id)
);
CREATE TABLE OrderContainsItem (
    order_id INTEGER NOT NULL,
    menu_item_id INTEGER NOT NULL,
    quantity INTEGER NOT NULL,
    PRIMARY KEY (order_id, menu_item_id),
    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id),
    FOREIGN KEY (menu_item_id)
        REFERENCES MenuItem(menu_item_id)
);
CREATE TABLE Reservation (
    reservation_id INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id INTEGER NOT NULL,
    restaurant_id INTEGER NOT NULL,
    table_number INTEGER,
    reservation_date TEXT NOT NULL,
    reservation_time TEXT NOT NULL,

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number_of_guests INTEGER,
reservation_status TEXT,
seating_start_time TEXT,
seating_end_time TEXT,
FOREIGN KEY (customer_id)
    REFERENCES Customer(customer_id),
FOREIGN KEY (restaurant_id)
    REFERENCES Restaurant(restaurant_id),
FOREIGN KEY (restaurant_id, table_number)
    REFERENCES DiningTable(restaurant_id, table_number)
);
COMMIT;

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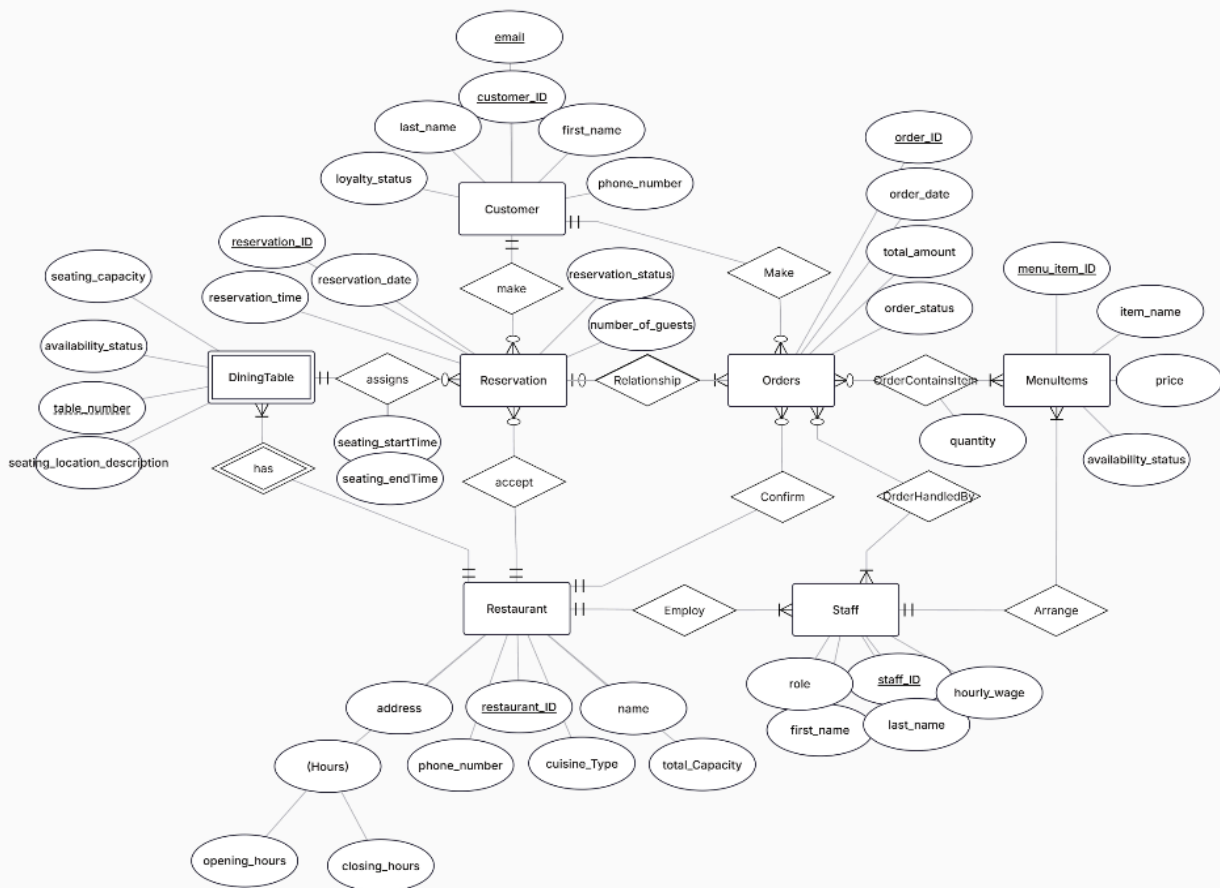


TABLE - Restaurant:

The restaurant table stores information about each restaurant, including its ID, location and details for its day to day operations. The primary key is restaurant_id and it's a strong entity, and one that has no foreign key but other entities rely on as their foreign key.

TABLE - Customer:

The customer table represents information about the individuals who place orders or make reservations at a restaurant. Each customer is uniquely identified given its primary key customer_id; It is a very strong entity, one which has no foreign keys but belongs as a foreign key for other entities.

TABLE - DiningTable:

The DiningTable table represents the actual physical tables within a restaurant and is clearly identified as the weak entity. It uses a partial key as table_number. Now since table_number is not sufficient or unique and can trace to any other restaurant, the table needs to be tied in with its singular strong entity relationship which is the restaurant, and its foreign key, restaurant_id. The Primary key for the DiningTable is (restaurant_id, table_number) which includes the partial key + foreign key to preserve identification dependency.

TABLE - Reservation:

The Reservation table stores booking information made by the customers for specific dates and times. It has its primary key as reservation_id and it has a lot of foreign keys coming from the Customer, Restaurant and the weak entity DiningTable that comes as a composite key.

TABLE - Orders:

The Orders table records any food orders placed by the customers. It includes not only reservation-based orders, but also walk-in orders made directly by the customer. It uses order_id as its primary key and includes foreign keys from Customer and Restaurant, and an optional foreign key with the Reservation(if the order comes from reservations)

TABLE - MenuItem:

The MenuItem table stores food and beverage items offered by a restaurant. The Items in the menu are identified uniquely and have as its primary key the menu_item_id attribute. It also includes foreign keys to the Staff which indicates which staff/manager has given permission to proceed with the item on the menu.

TABLE - OrderContainsItem:

The OrderContainsItem table represents the many-to-many relationship between Orders and the menu items. It uses a composite primary key that is made up of foreign keys of both the Orders and MenuItem, as (order_id, menu_item_id) along with the "relationship" attribute of quantity.

TABLE - Staff:

The Staff table stores information about employees that work at the restaurant. The primary unique key is the staff_id and the foreign key comes from a Restaurant indicating the workplace that each staff member works at.

TABLE - OrderHandledBy:

The OrderHandledBy table models the many-to-many relationship between staff members and the orders

they have handled. The table has a composite primary key made from foreign keys referencing the Orders and Staff tables, (order_id, staff_id).